Document Investigation Dossier

Case **DEMO-TAMPERED** · generated 06 Nov 2026, 17:37 · analyst: Ayush Kumar

4	3	5	9	HIGH
documents	potential manipulation	inconsistencies	observations	review priority

Executive summary

4 documents were examined (Identity Card, Salary Slip, Loan Application Form, Bank Statement). The system recorded 3 potential manipulation hypotheses, 5 detected inconsistencies and 9 observations. Review priority: HIGH. Fields with converging evidence of alteration: Salary Slip - Net pay; Loan Application Form - Monthly net income; Bank Statement - Closing balance. Cross-document conflicts: Date of birth mismatch: Loan Application Form says '14/03/1994', Identity Card says '14/03/1996'; Monthly net income mismatch: Loan Application Form says '95,000', Bank Statement says '45,200'; Monthly net income mismatch: Salary Slip says '95,200', Bank Statement says '45,200'. The applicant's stated income (95,000) agrees with the payslip Net Pay (95,200), but that payslip value is itself under suspicion, so this agreement is not treated as independent corroboration. All findings are indicators for human review. None of them is a determination of fraud.

Status definitions. Observation: a visual or technical characteristic was observed; on its own it is not a problem. **Detected inconsistency**: a verifiable conflict in the data (arithmetic or between documents). **Potential manipulation**: several independent signals converge on the same field. None of these is a finding of fraud; consequential decisions remain with a human reviewer.

7.1 Document overview

ID	File	Type (confidence)	Source	Pages	Status	Review priority	SHA-256 (first 16)
DOC-01	01_identity_card.jpg	Identity Card (99%)	scanned (OCR)	1	investigated	MEDIUM (0.56)	4023d2cfd6352213
DOC-02	02_salary_slip.jpg	Salary Slip (99%)	scanned (OCR)	1	investigated	HIGH (0.9)	f32299ad1d65f2a3
DOC-03	03_loan_application.jpg	Loan Application Form (99%)	scanned (OCR)	1	investigated	HIGH (0.92)	0fda9e2cd8ed5daf
DOC-04	04_bank_statement.pdf	Bank Statement (99%)	digital PDF	1	investigated	HIGH (0.94)	913ebc65bbbfda68

Full SHA-256 hashes are recorded in the audit trail at ingestion so the analysed files can be matched to the originals (ISO/IEC 27037 practice).

7.2 Extracted information

Every value keeps its page and bounding box (x0, y0, x1, y1 in 150-dpi pixels) so it can be traced to its location in the original document. Confidence is the OCR engine's word confidence; for digital PDFs the text layer is exact (99).

DOC-01 · Identity Card · 01_identity_card.jpg

Field	Value (normalised)	As read	Conf.	Page	Location
Name	Rohan Verma	Rohan Verma	96	1	(373, 265, 554, 286)
Date of birth	14/03/1996	14/03/1996	96	1	(373, 336, 530, 359)
ID number	DEMO-4821-7730-15	DEMO-4821-7730-15	92	1	(373, 406, 662, 426)
Gender	Male	Male	96	1	(373, 475, 435, 496)
Address	21 Lake View Road, Bhopal 462001	21 Lake View Road, Bhopal 462001	96	1	(372, 545, 862, 572)

Structure: completeness 100%; missing: none; duplicated: none; value columns at x = [373].

DOC-02 · Salary Slip · 02_salary_slip.jpg

Field	Value (normalised)	As read	Conf.	Page	Location
Name	Rohan Verma	Rohan Verma	96	1	(413, 285, 594, 306)
Employee ID	NSW-2291	NSW-2291	92	1	(413, 346, 554, 366)
Pay period	September 2026	September 2026	96	1	(412, 405, 643, 432)
Pay date	30/09/2026	30/09/2026	96	1	(412, 466, 570, 489)
Basic	30,000	30,000	96	1	(472, 611, 559, 633)
Provident fund	3,600	3,600	96	1	(1042, 611, 1113, 633)
HRA	12,000	12,000	96	1	(473, 661, 559, 683)
Professional tax	200	200	96	1	(1042, 661, 1088, 680)
Special allowance	10,000	10,000	96	1	(473, 711, 559, 733)
Income tax	3,000	3,000	96	1	(1042, 711, 1113, 733)
Gross earnings	52,000	52,000	96	1	(412, 786, 506, 810)
Total deductions	6,800	6,800	96	1	(412, 846, 488, 870)
Net pay	95,200	95,200	96	1	(414, 914, 519, 942)
Employer	Nimbus Softworks Pvt Ltd	Nimbus Softworks Pvt Ltd	96	1	(84, 78, 663, 108)

Structure: completeness 100%; missing: none; duplicated: none; value columns at x = [413, 473, 1042].

DOC-03 · Loan Application Form · 03_loan_application.jpg

Field	Value (normalised)	As read	Conf.	Page	Location
Name	Rohan Verma	Rohan Verma	96	1	(463, 215, 644, 236)
Date of birth	14/03/1994	14/03/1994	95	1	(463, 291, 620, 314)
ID number	DEMO-4821-7730-15	DEMO-4821-7730-15	92	1	(463, 366, 752, 386)
Address	21 Lake View Road, Bhopal 462001	21 Lake View Road, Bhopal 462001	96	1	(462, 440, 952, 467)

Field	Value (normalised)	As read	Conf.	Page	Location
Employer	Nimbus Softworks Pvt Ltd	Nimbus Softworks Pvt Ltd	96	1	(463, 515, 817, 536)
Monthly net income	95,000	95,000	96	1	(464, 589, 569, 617)
Loan amount	300,000	3,00,000	91	1	(462, 666, 583, 690)
Loan tenure	36 months	36 months	95	1	(462, 740, 608, 761)
Application date	05/10/2026	05/10/2026	96	1	(462, 816, 620, 839)

Structure: completeness 100%; missing: none; duplicated: none; value columns at x = [463].

DOC-04 · Bank Statement · 04_bank_statement.pdf

Field	Value (normalised)	As read	Conf.	Page	Location
Name	Rohan Verma	Rohan Verma	99	1	(395, 204, 537, 236)
Account number	XXXXXXXX5521	XXXXXXXX5521	99	1	(395, 242, 569, 273)
Address	21 Lake View Road, Bhopal 462001	21 Lake View Road, Bhopal 462001	99	1	(395, 279, 761, 311)
Statement period	01/09/2026 to 30/09/2026	01/09/2026 to 30/09/2026	99	1	(395, 317, 657, 348)
Closing balance	99,220	99,220.00	99	1	(395, 640, 491, 671)

Structure: completeness 100%; missing: none; duplicated: none; value columns at x = [396].

Transactions table (structured)

Date	Description	Debit	Credit	Balance
01/09/2026	Opening Balance	-	-	24,450
03/09/2026	UPI GROCERY MART	3,250	-	21,200
05/09/2026	RENT TRANSFER	15,000	-	6,200
12/09/2026	ELECTRICITY BILL	1,180	-	5,020
30/09/2026	SALARY NIMBUS SOFTWORKS	-	45,200	50,220

7.3 Contextual analysis

Cross-document entity matrix

Each attribute is compared with a reference source (identity attributes: identity card; income: bank statement, as an independent third-party record). ✓ = consistent, ✗ = conflict, REF = reference.

Attribute	Reference	Values found (document: value → result)
Name	Identity Card	DOC-01 Identity Card: **Rohan Verma** → REF DOC-02 Salary Slip: **Rohan Verma** → OK DOC-03 Loan Application Form: **Rohan Verma** → OK DOC-04 Bank Statement: **Rohan Verma** → OK
Date of birth	Identity Card	DOC-01 Identity Card: **14/03/1996** → REF DOC-03 Loan Application Form: **14/03/1994** → CONFLICT (score 90.0)
ID number	Identity Card	DOC-01 Identity Card: **DEMO-4821-7730-15** → REF DOC-03 Loan Application Form: **DEMO-4821-7730-15** → OK
Address	Identity Card	DOC-01 Identity Card: **21 Lake View Road, Bhopal 462001** → REF DOC-03 Loan Application Form: **21 Lake View Road, Bhopal 462001** → OK DOC-04 Bank Statement: **21 Lake View Road, Bhopal 462001** → OK
Employer	Salary Slip	DOC-02 Salary Slip: **Nimbus Softworks Pvt Ltd** → REF DOC-03 Loan Application Form: **Nimbus Softworks Pvt Ltd** → OK DOC-04 Bank Statement: **SALARY NIMBUS SOFTWORKS** → OK
Monthly net income	Bank Statement	DOC-03 Loan Application Form: **95,000** → CONFLICT (score 47.6) DOC-02 Salary Slip: **95,200** → CONFLICT (score 47.5) DOC-04 Bank Statement: **45,200** → REF

Inconsistencies within and across documents

ID	Finding	Status	Sev.	Conf.	Related
F-011	Bank Statement: Closing Balance does not match the last running balance	Detected inconsistency	high	94%	F-007, F-008, F-017
F-010	Salary Slip: Net Pay does not equal Gross minus Deductions	Detected inconsistency	high	91%	F-002, F-003, F-014, F-015
F-013	Monthly net income mismatch: Loan Application Form says '95,000', Bank Statement says '45,200'	Detected inconsistency	high	88%	F-004, F-005, F-016
F-014	Monthly net income mismatch: Salary Slip says '95,200', Bank Statement says '45,200'	Detected inconsistency	high	88%	F-002, F-003, F-010, F-015
F-012	Date of birth mismatch: Loan Application Form says '14/03/1994', Identity Card says '14/03/1996'	Detected inconsistency	high	87%	-

Reasoning note: The applicant's stated income (95,000) agrees with the payslip Net Pay (95,200), but that payslip value is itself under suspicion, so this agreement is not treated as independent corroboration.

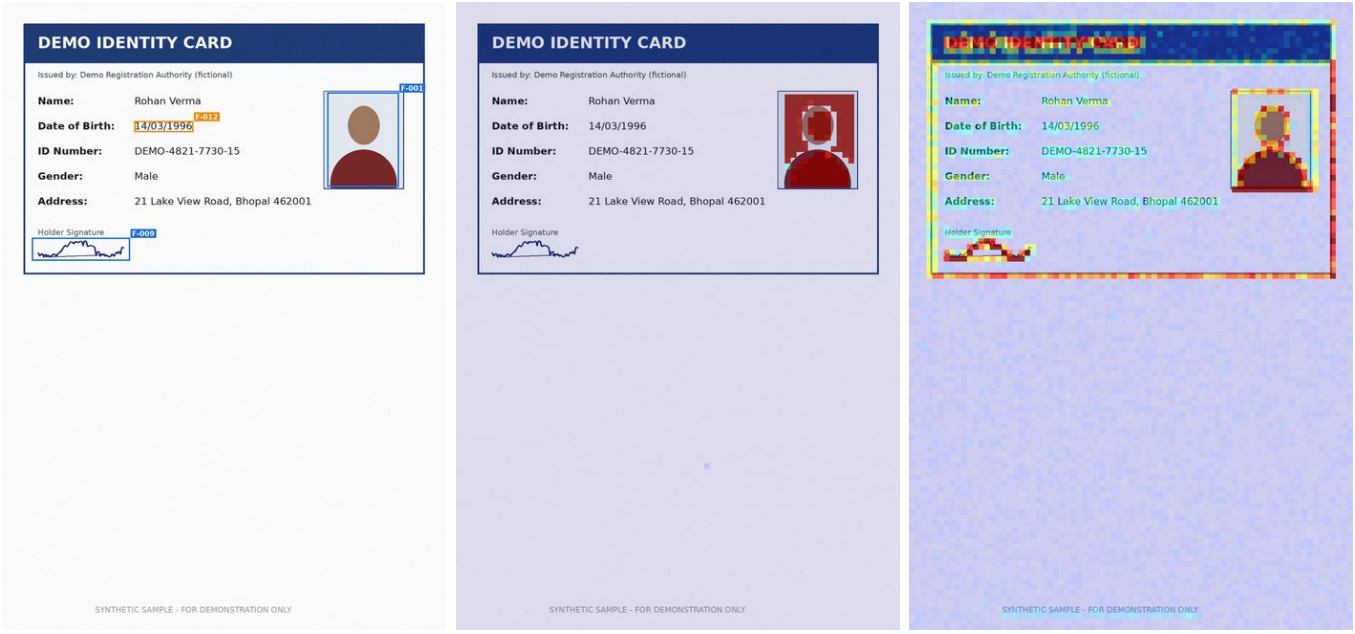
7.4 Visual forensic findings

ID	Finding	Status	Sev.	Conf.	Related
F-001	Identity Card: region lacks scanner noise near an embedded image / photograph area	Observation	high	90%	-
F-002	Salary Slip: region lacks scanner noise near the value of 'Net pay'	Observation	high	90%	F-003, F-010, F-014, F-015
F-004	Loan Application Form: region lacks scanner noise near the value of 'Monthly net income'	Observation	high	90%	F-005, F-013, F-016
F-007	Bank Statement: 'Closing balance' uses a font foreign to the rest of the PDF	Observation	high	85%	F-011, F-008, F-017
F-003	Salary Slip: character size of 'Net pay' differs from aligned values	Observation	medium	80%	F-002, F-010, F-014, F-015
F-005	Loan Application Form: character size of 'Monthly net income' differs from aligned values	Observation	medium	80%	F-004, F-013, F-016
F-006	Loan Application Form: signature placed outside the designated signature area	Observation	medium	80%	-
F-008	Bank Statement: PDF metadata indicates editing after creation	Observation	medium	80%	F-017
F-009	Signature comparison Identity Card vs Loan Application Form: visually consistent (score 0.98)	Observation	low	50%	-

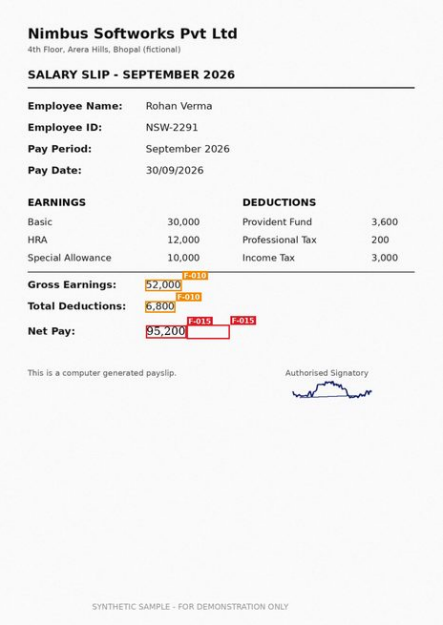
Annotated pages and forensic maps

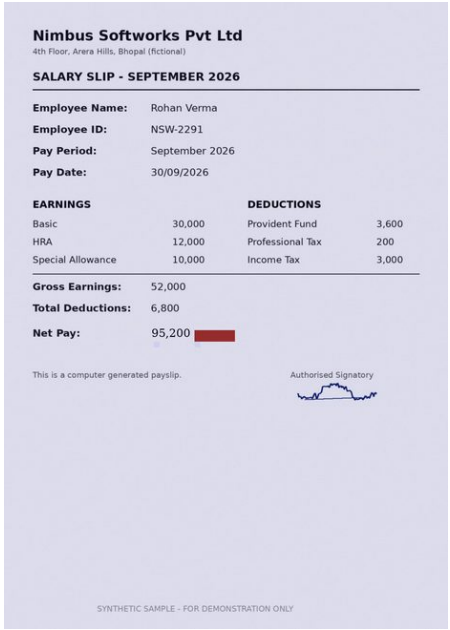
Left: findings drawn on the page (red = potential manipulation, orange = inconsistency, blue = observation). Middle: noise-consistency map (bright = abnormally clean). Right: Error Level Analysis.

DOC-01 · Identity Card



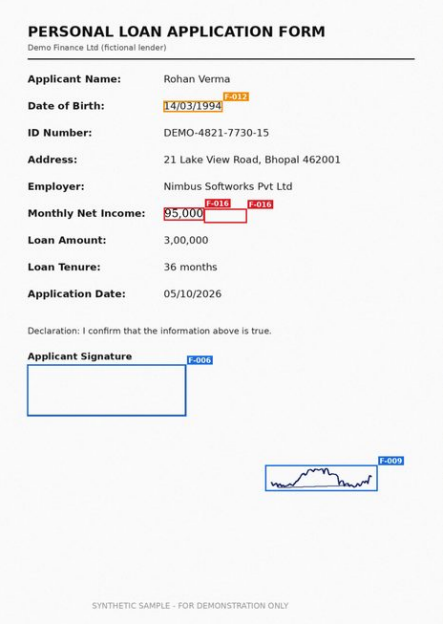
DOC-02 · Salary Slip







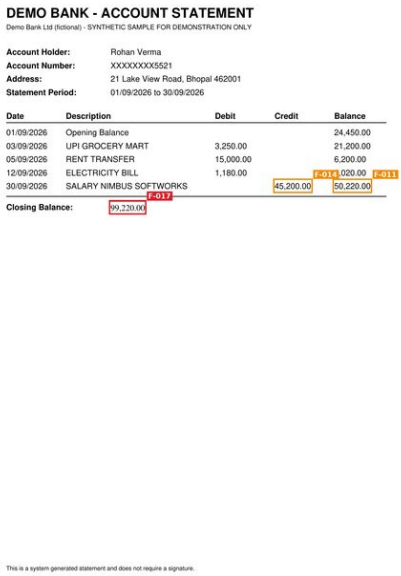
DOC-03 · Loan Application Form







DOC-04 · Bank Statement



DEMO BANK - ACCOUNT STATEMENT

Demo Bank Ltd (fictional) - SYNTHETIC SAMPLE FOR DEMONSTRATION ONLY

Account Holder: Rohan Verma
Account Number: XXXXXXXX5521
Address: 21 Lake View Road, Bhopal 462001
Statement Period: 01/09/2026 to 30/09/2026

Date	Description	Debit	Credit	Balance
01/09/2026	Opening Balance			24,450.00
03/09/2026	UPI GROCERY MART	3,250.00		21,200.00
05/09/2026	RENT TRANSFER	15,000.00		6,200.00
12/09/2026	ELECTRICITY BILL	1,180.00		5,020.00
30/09/2026	SALARY NIMBUS SOFTWORKS		45,200.00	50,220.00

Closing Balance: 99,220.00

This is a system generated statement and does not require a signature.

digital PDF: noise analysis not applicable; text-layer font and metadata analysis used instead

7.5 Evidence and traceability

For each finding: the evidence items, their source document, page and bounding box, and an image crop of the original region. Related findings are cross-referenced.

F-015 **Salary Slip: 'Net pay' shows converging signs of alteration (stated value 95,200)** **Potential manipulation (needs verification)** **HIGH · 97%**

4 findings from independent visual/digital (2) and data-consistency (2) checks all concern the same field. Visual traces show how it may have been changed; data conflicts show that the value disagrees with the rest of the submission. Converging, independent signals are much stronger than any single one. Independent sources point to 45,200 as the consistent value (recomputed from components (F-010); Bank Statement (F-014)), whereas the document states 95,200.
- [F-002] Salary Slip: region lacks scanner noise near the value of 'Net pay'
- [F-003] Salary Slip: character size of 'Net pay' differs from aligned values
- [F-010] Salary Slip: Net Pay does not equal Gross minus Deductions
- [F-014] Monthly net income mismatch: Salary Slip says '95,200', Bank Statement says '45,200'

Doc	Page	Location	Field / value	Note	Crop
DOC-02	1	(414, 914, 519, 942)	Net pay: 95,200	field under investigation	
DOC-02	1	(528, 912, 640, 944)	region	noise level 0.03 vs page 1.4 (x0.02); ELA response x0.0	

Verify: Treat 'Net pay' as unverified. Obtain the original from the issuer and confirm whether 45,200 is correct. This is a hypothesis for human review, not a determination of fraud. · **Rule:** R1-convergence · **Related:** F-002, F-003, F-010, F-014

F-016 **Loan Application Form: 'Monthly net income' shows converging signs of alteration (stated value 95,000)** **Potential manipulation (needs verification)** **HIGH · 97%**

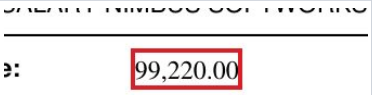
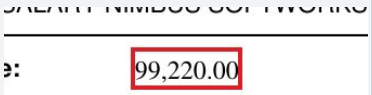
3 findings from independent visual/digital (2) and data-consistency (1) checks all concern the same field. Visual traces show how it may have been changed; data conflicts show that the value disagrees with the rest of the submission. Converging, independent signals are much stronger than any single one. Independent sources point to 45,200 as the consistent value (Bank Statement (F-013)), whereas the document states 95,000.
- [F-004] Loan Application Form: region lacks scanner noise near the value of 'Monthly net income'
- [F-005] Loan Application Form: character size of 'Monthly net income' differs from aligned values
- [F-013] Monthly net income mismatch: Loan Application Form says '95,000', Bank Statement says '45,200'

Doc	Page	Location	Field / value	Note	Crop
DOC-03	1	(464, 589, 569, 617)	Monthly net income: 95,000	field under investigation	
DOC-03	1	(576, 592, 688, 624)	region	noise level 0.04 vs page 1.4 (x0.03); ELA response x0.0	

Verify: Treat 'Monthly net income' as unverified. Obtain the original from the issuer and confirm whether 45,200 is correct. This is a hypothesis for human review, not a determination of fraud. · **Rule:** R1-convergence · **Related:** F-004, F-005, F-013

F-017 **Bank Statement: 'Closing balance' shows converging signs of alteration (stated value 99,220.00)** Potential manipulation (needs verification) HIGH · 97%

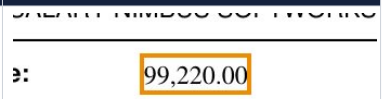
3 findings from independent visual/digital (2) and data-consistency (1) checks all concern the same field. Visual traces show how it may have been changed; data conflicts show that the value disagrees with the rest of the submission. Converging, independent signals are much stronger than any single one. Independent sources point to 50,220 as the consistent value (recomputed from components (F-011)), whereas the document states 99,220.00.
- [F-007] Bank Statement: 'Closing balance' uses a font foreign to the rest of the PDF
- [F-011] Bank Statement: Closing Balance does not match the last running balance
- [F-008] Bank Statement: PDF metadata indicates editing after creation

Doc	Page	Location	Field / value	Note	Crop
DOC-04	1	(395, 640, 491, 671)	Closing balance: 99,220.00	field under investigation	
DOC-04	1	(395, 639, 491, 671)	value: 99,220.00	font Times-Roman (rest of document: helvetica)	
DOC-04	-	-	metadata	Created by 'DemoBank eStatement v4' but last written by 'QuickPDF Editor 2.1'	-
DOC-04	-	-	metadata	Modified (04/10/2026 21:30) after creation (30/09/2026 18:00)	-

Verify: Treat 'Closing balance' as unverified. Obtain the original from the issuer and confirm whether 50,220 is correct. This is a hypothesis for human review, not a determination of fraud. · **Rule:** R1-convergence · **Related:** F-007, F-011, F-008

F-011 **Bank Statement: Closing Balance does not match the last running balance** Detected inconsistency HIGH · 94%

The transaction table ends at 50,220 but the Closing Balance line reads 99,220 (difference 49,000). Every transaction was re-added; the table itself is internally consistent, so the summary line is the outlier.

Doc	Page	Location	Field / value	Note	Crop
DOC-04	1	(395, 640, 491, 671)	Closing balance: 99,220.00	stated closing balance	
DOC-04	1	(1020, 579, 1122, 611)	value: 50,220	last running balance in table	

Verify: Request the statement from the bank. · **Rule:** C5-bank-closing · **Related:** F-007, F-008, F-017

F-010 **Salary Slip: Net Pay does not equal Gross minus Deductions** Detected inconsistency HIGH · 91%

Gross Earnings - Total Deductions = 45,200, but Net Pay reads 95,200 (difference 50,000). Payroll systems compute Net Pay automatically, so an arithmetic break points to the stated figure having been changed - or an OCR misread.

Doc	Page	Location	Field / value	Note	Crop
DOC-02	1	(412, 786, 506, 810)	Gross earnings: 52,000	Gross earnings as read (OCR conf 96%)	
DOC-02	1	(412, 846, 488, 870)	Total deductions: 6,800	Total deductions as read (OCR conf 96%)	
DOC-02	1	(414, 914, 519, 942)	Net pay: 95,200	stated total	
DOC-02	-	-	value: 45,200	recomputed from components	-

Verify: Recompute from the source system; ask the issuer which figure is correct. · **Rule:** C3-salary-net · **Related:** F-002, F-003, F-014, F-015

F-013 **Monthly net income mismatch: Loan Application Form says '95,000', Bank Statement says '45,200'** Detected inconsistency HIGH · 88%

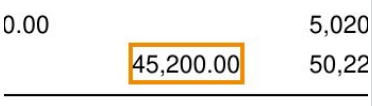
Income stated on an application should be supported by payslip and bank credit. A mismatch is one of the strongest indicators of an inflated income claim. Similarity score 47.6 (reference document: Bank Statement).

Doc	Page	Location	Field / value	Note	Crop
DOC-04	1	(854, 579, 956, 611)	salary_credit: 45,200	reference value	
DOC-03	1	(464, 589, 569, 617)	Monthly net income: 95,000	conflicting value	

Verify: Ask the applicant to explain the difference; verify with the issuer of the reference document. · **Rule:** X-income · **Related:** F-004, F-005, F-016

F-014 **Monthly net income mismatch: Salary Slip says '95,200', Bank Statement says '45,200'** Detected inconsistency HIGH · 88%

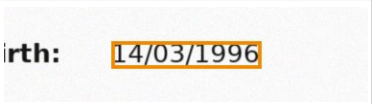
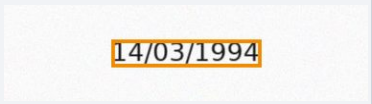
Income stated on an application should be supported by payslip and bank credit. A mismatch is one of the strongest indicators of an inflated income claim. Similarity score 47.5 (reference document: Bank Statement).

Doc	Page	Location	Field / value	Note	Crop
DOC-04	1	(854, 579, 956, 611)	salary_credit: 45,200	reference value	
DOC-02	1	(414, 914, 519, 942)	Net pay: 95,200	conflicting value	

Verify: Ask the applicant to explain the difference; verify with the issuer of the reference document. · **Rule:** X-income · **Related:** F-002, F-003, F-010, F-015

F-012 **Date of birth mismatch: Loan Application Form says '14/03/1994', Identity Card says '14/03/1996'** Detected inconsistency **HIGH · 87%**

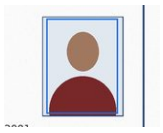
Date of birth is a stable identity attribute; documents for the same person must agree. Similarity score 90.0 (reference document: Identity Card).

Doc	Page	Location	Field / value	Note	Crop
DOC-01	1	(373, 336, 530, 359)	Date of birth: 14/03/1996	reference value	
DOC-03	1	(463, 291, 620, 314)	Date of birth: 14/03/1994	conflicting value	

Verify: Ask the applicant to explain the difference; verify with the issuer of the reference document. · **Rule:** X-dob

F-001 **Identity Card: region lacks scanner noise near an embedded image / photograph area** Observation **HIGH · 90%**

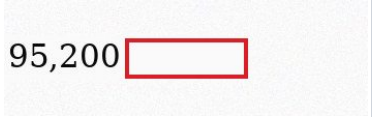
Scanned paper carries roughly uniform sensor noise (page median 1.4). A 192x256 px area near an embedded image / photograph area is almost noise-free (x0.0). Areas added digitally after scanning, such as a pasted patch or replaced photo, typically look like this.

Doc	Page	Location	Field / value	Note	Crop
DOC-01	1	(912, 256, 1104, 512)	region	noise level 0.0 vs page 1.4 (x0.0); ELA response x1.24	

Verify: Compare with the physical original or an independently obtained copy. · **Rule:** V1-noise-consistency

F-002 **Salary Slip: region lacks scanner noise near the value of 'Net pay'** Observation **HIGH · 90%**

Scanned paper carries roughly uniform sensor noise (page median 1.4). A 112x32 px area near the value of 'Net pay' is almost noise-free (x0.02). Areas added digitally after scanning, such as a pasted patch or replaced photo, typically look like this. Error Level Analysis also responds differently here (x0.0 of the page average), suggesting a different compression history.

Doc	Page	Location	Field / value	Note	Crop
DOC-02	1	(528, 912, 640, 944)	region	noise level 0.03 vs page 1.4 (x0.02); ELA response x0.0	
DOC-02	1	(414, 914, 519, 942)	Net pay: 95,200	field adjacent to the clean region	

Verify: Compare with the physical original or an independently obtained copy. · **Rule:** V1-noise-consistency · **Related:** F-003, F-010, F-014, F-015

F-004 **Loan Application Form: region lacks scanner noise near the value of 'Monthly net income'** Observation **HIGH · 90%**

Scanned paper carries roughly uniform sensor noise (page median 1.4). A 112x32 px area near the value of 'Monthly net income' is almost noise-free (x0.03). Areas added digitally after scanning, such as a pasted patch or replaced photo, typically look like this. Error Level Analysis also responds differently here (x0.0 of the page average), suggesting a different compression history.

Doc	Page	Location	Field / value	Note	Crop
DOC-03	1	(576, 592, 688, 624)	region	noise level 0.04 vs page 1.4 (x0.03); ELA response x0.0	
DOC-03	1	(464, 589, 569, 617)	Monthly net income: 95,000	field adjacent to the clean region	

Verify: Compare with the physical original or an independently obtained copy. · **Rule:** V1-noise-consistency · **Related:** F-005, F-013, F-016

F-007 **Bank Statement: 'Closing balance' uses a font foreign to the rest of the PDF** Observation **HIGH · 85%**

The PDF text layer shows '99,220.00' rendered in Times-Roman, while the rest of the document uses helvetica (1.3% of words use this font). Statement generators use a fixed font set; a single foreign-font value is typical of an edit made in a PDF editor.

Doc	Page	Location	Field / value	Note	Crop
DOC-04	1	(395, 639, 491, 671)	value: 99,220.00	font Times-Roman (rest of document: helvetica)	

Verify: Obtain the statement directly from the issuing bank. · **Rule:** D1-pdf-font · **Related:** F-011, F-008, F-017

F-003 **Salary Slip: character size of 'Net pay' differs from aligned values** Observation **MEDIUM · 80%**

Values in the same column (Pay date, Gross earnings, Total deductions) have a median character height of 20px; this value measures 23px (+15%). Documents generated by one system usually use one font size per column, so a single outlier suggests the value was typed separately.

Doc	Page	Location	Field / value	Note	Crop
DOC-02	1	(414, 914, 519, 942)	Net pay: 95,200	glyph height 23px vs column 20px	

Verify: Check the typeface and size of this value against the original. · **Rule:** V3-glyph-size · **Related:** F-002, F-010, F-014, F-015

F-005 **Loan Application Form: character size of 'Monthly net income' differs from aligned values** Observation **MEDIUM · 80%**

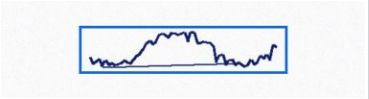
Values in the same column (Loan amount, Application date, Date of birth, ID number) have a median character height of 20px; this value measures 23px (+15%). Documents generated by one system usually use one font size per column, so a single outlier suggests the value was typed separately.

Doc	Page	Location	Field / value	Note	Crop
DOC-03	1	(464, 589, 569, 617)	Monthly net income: 95,000	glyph height 23px vs column 20px	

Verify: Check the typeface and size of this value against the original. · **Rule:** V3-glyph-size · **Related:** F-004, F-013, F-016

F-006 **Loan Application Form: signature placed outside the designated signature area** Observation **MEDIUM · 80%**

A signature-like ink stroke was found at (747, 1308, 1053, 1373), but the designated drawn box at (82, 1027, 519, 1164) is empty. A signature outside its box can indicate it was transplanted from another document.

Doc	Page	Location	Field / value	Note	Crop
DOC-03	1	(747, 1308, 1053, 1373)	region	detected signature	
DOC-03	1	(82, 1027, 519, 1164)	region	designated signature area (empty)	

Verify: Ask for a freshly signed copy; compare with the signature on file. · **Rule:** V4-signature-placement

F-008 **Bank Statement: PDF metadata indicates editing after creation** Observation **MEDIUM · 80%**

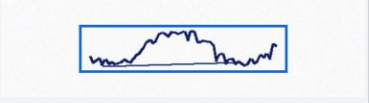
Created by 'DemoBank eStatement v4' but last written by 'QuickPDF Editor 2.1'; Modified (04/10/2026 21:30) after creation (30/09/2026 18:00). Metadata alone is weak evidence (it can be changed legitimately or forged), but it supports other signals on this file.

Doc	Page	Location	Field / value	Note	Crop
DOC-04	-	-	metadata	Created by 'DemoBank eStatement v4' but last written by 'QuickPDF Editor 2.1'	-
DOC-04	-	-	metadata	Modified (04/10/2026 21:30) after creation (30/09/2026 18:00)	-

Verify: Request the original file from the issuer. · **Rule:** D2-pdf-metadata · **Related:** F-017

F-009 **Signature comparison Identity Card vs Loan Application Form: visually consistent (score 0.98)** Observation **LOW · 50%**

Shape correlation of the two signature ink masks is 0.98 (1.0 = identical shape). This is a coarse shape heuristic, not handwriting verification: it cannot authenticate a signer and a high score can also mean a signature was copied.

Doc	Page	Location	Field / value	Note	Crop
DOC-01	1	(87, 662, 353, 718)	region	signature A	
DOC-03	1	(747, 1308, 1053, 1373)	region	signature B	

Verify: Forensic handwriting examination if the signature matters to the decision. · **Rule:** V4-signature-compare

7.6 Investigation summary

Identified anomalies (ranked)

ID	Anomaly	Status	Conf.
F-015	Salary Slip: 'Net pay' shows converging signs of alteration (stated value 95,200)	Potential manipulation (needs verification)	97%
F-016	Loan Application Form: 'Monthly net income' shows converging signs of alteration (stated value 95,000)	Potential manipulation (needs verification)	97%
F-017	Bank Statement: 'Closing balance' shows converging signs of alteration (stated value 99,220.00)	Potential manipulation (needs verification)	97%
F-011	Bank Statement: Closing Balance does not match the last running balance	Detected inconsistency	94%
F-010	Salary Slip: Net Pay does not equal Gross minus Deductions	Detected inconsistency	91%
F-013	Monthly net income mismatch: Loan Application Form says '95,000', Bank Statement says '45,200'	Detected inconsistency	88%
F-014	Monthly net income mismatch: Salary Slip says '95,200', Bank Statement says '45,200'	Detected inconsistency	88%
F-012	Date of birth mismatch: Loan Application Form says '14/03/1994', Identity Card says '14/03/1996'	Detected inconsistency	87%
F-001	Identity Card: region lacks scanner noise near an embedded image / photograph area	Observation	90%
F-002	Salary Slip: region lacks scanner noise near the value of 'Net pay'	Observation	90%
F-004	Loan Application Form: region lacks scanner noise near the value of 'Monthly net income'	Observation	90%
F-007	Bank Statement: 'Closing balance' uses a font foreign to the rest of the PDF	Observation	85%
F-003	Salary Slip: character size of 'Net pay' differs from aligned values	Observation	80%
F-005	Loan Application Form: character size of 'Monthly net income' differs from aligned values	Observation	80%
F-006	Loan Application Form: signature placed outside the designated signature area	Observation	80%
F-008	Bank Statement: PDF metadata indicates editing after creation	Observation	80%

Potential explanations

F-015 - Salary Slip: 'Net pay' shows converging signs of alteration (stated value 95,200). Possible explanations, in the order to rule out: (1) OCR misread of the value (check the crop); (2) a legitimate correction or re-issue by the issuer; (3) alteration after issue. Independent sources agree on 45,200.

F-016 - Loan Application Form: 'Monthly net income' shows converging signs of alteration (stated value 95,000). Possible explanations, in the order to rule out: (1) OCR misread of the value (check the crop); (2) a legitimate correction or re-issue by the issuer; (3) alteration after issue. Independent sources agree on 45,200.

F-017 - Bank Statement: 'Closing balance' shows converging signs of alteration (stated value 99,220.00). Possible explanations, in the order to rule out: (1) OCR misread of the value (check the crop); (2) a legitimate correction or re-issue by the issuer; (3) alteration after issue. Independent sources agree on 50,220.

Areas requiring additional verification

• Treat 'Net pay' as unverified. Obtain the original from the issuer and confirm whether 45,200 is correct. This is a hypothesis for human review, not a determination of fraud. (F-015)

• Treat 'Monthly net income' as unverified. Obtain the original from the issuer and confirm whether 45,200 is correct. This is a hypothesis for human review, not a determination of fraud. (F-016)

• Treat 'Closing balance' as unverified. Obtain the original from the issuer and confirm whether 50,220 is correct. This is a hypothesis for human review, not a determination of fraud. (F-017)

• Request the statement from the bank. (F-011)

- Recompute from the source system; ask the issuer which figure is correct. (F-010)
- Ask the applicant to explain the difference; verify with the issuer of the reference document. (F-013)
- Compare with the physical original or an independently obtained copy. (F-001)
- Obtain the statement directly from the issuing bank. (F-007)
- Check the typeface and size of this value against the original. (F-003)
- Ask for a freshly signed copy; compare with the signature on file. (F-006)
- Request the original file from the issuer. (F-008)

Investigation limitations

- Prototype rules and layouts were developed on the synthetic demonstration dataset; new document templates need schema entries.
- Noise-consistency analysis assumes the page was scanned or photographed. Born-digital images have little noise and are analysed with lower sensitivity; a forger who adds matching noise can evade it.
- Glyph-size checks need at least three aligned numeric values in a column.
- Signature detection currently finds coloured (blue) ink only; black-ink signatures may be missed. Signature comparison is a coarse shape heuristic, not handwriting verification.
- Handwritten field recognition is not supported in this version; multilingual OCR is possible through Tesseract language packs but not tested.
- Cross-document reference choices (ID card for identity, bank statement for income) are assumptions; a manipulated reference document weakens the comparison, which is why visual and digital checks run on every document.
- OCR errors can create false inconsistencies; every data finding is scaled by the OCR confidence of the fields it uses, and crops are provided so a reviewer can read the value directly.
- PDF metadata can be altered or legitimately changed; it is used only as supporting evidence.

Appendix A Methodology and technologies

Stage	Technique	Implementation
Ingestion	SHA-256 hashing, page rendering at 150 dpi	Pillow, PyMuPDF
Classification	Keyword-evidence scoring with confidence margin	Python
OCR & extraction	Tesseract LSTM OCR; row/segment layout grouping; label→value pairing with fuzzy label matching; table parsing by column geometry	Tesseract 5, pytesseract, RapidFuzz
Structure	Expected-field schema per document type; missing / duplicate / low-confidence detection; value-column alignment	Python
Visual forensics	Noise-residual consistency (block-wise), Error Level Analysis, glyph-height outliers per value column, coloured-ink signature detection, box detection, signature shape correlation	OpenCV, NumPy
Digital forensics	PDF text-layer font census; metadata (producer vs creator, modification date, incremental saves)	PyMuPDF
Contextual verification	Arithmetic rules (payslip totals, bank running balance), date and age logic	Python rules
Cross-document	Entity matrix with reference sources; fuzzy name/address matching; tolerance-based amount matching; temporal checks	RapidFuzz
Reasoning	Evidence graph keyed by (document, field); convergence of independent signal types; Bayesian-style confidence combination; likely-original estimation; discounting of suspicious corroboration	Python
Narrative (optional)	LLM rewrites the deterministic summary using only structured findings	Claude API (optional)
Evidence & audit	Region crops, annotated pages, hash-chained JSONL audit log, analyst review decisions	Pillow, hashlib

Reference frameworks used as guidance (no formal compliance claimed): ISO/IEC 27037, 27041, 27042; NIST SP 800-86; OWASP ML Security Top 10.

Appendix B Audit trail

Hash-chain integrity: **VERIFIED** - 30 entries verified. Each entry stores the SHA-256 of the previous entry; editing any entry breaks the chain.

#	Time (UTC)	Actor	Event	Details	Hash
1	17:37:23.555	docsentinel	case_opened	case_id=DEMO-TAMPERED, files=['01_identity_card.jpg', '02_salary_slip, analyst=Ayush Kumar	2355ab8ac6
2	17:37:23.995	docsentinel	document_ingested	doc_id=DOC-01, filename=01_identity_card.jpg, sha256=4023d2cfd63522136018a6413d9af848b4d1296b, media_type=image/jpg, pages=1, source=ocr	da432658ea
3	17:37:24.455	docsentinel	document_ingested	doc_id=DOC-02, filename=02_salary_slip.jpg, sha256=f32299ad1d65f2a3d0a5ddb0e3a23eab45b6cc41, media_type=image/jpg, pages=1, source=ocr	cdbcd4d2d0
4	17:37:24.856	docsentinel	document_ingested	doc_id=DOC-03, filename=03_loan_application.jpg, sha256=0fda9e2cd8ed5dafcb462d699ca06aa4b89a9958, media_type=image/jpg, pages=1, source=ocr	c3e3f84571
5	17:37:27.113	docsentinel	document_ingested	doc_id=DOC-04, filename=04_bank_statement.pdf, sha256=913ebc65bbbfda68cfdf0aa50ada4dc7a034e05c, media_type=application/pdf, pages=1, source=text-layer	617625046e
6	17:37:27.114	docsentinel	document_classified	doc_id=DOC-01, doc_type=identity_card, confidence=0.99	2c0606235e
7	17:37:27.114	docsentinel	document_classified	doc_id=DOC-02, doc_type=salary_slip, confidence=0.99	4832ae1923
8	17:37:27.114	docsentinel	document_classified	doc_id=DOC-03, doc_type=loan_application, confidence=0.99	20b1ac1cec
9	17:37:27.114	docsentinel	document_classified	doc_id=DOC-04, doc_type=bank_statement, confidence=0.99	a687ecde86
10	17:37:27.116	docsentinel	fields_extracted	doc_id=DOC-01, n_fields=5, mean_conf=95.2	2b29724fde
11	17:37:27.121	docsentinel	fields_extracted	doc_id=DOC-02, n_fields=14, mean_conf=95.7	39edf1b9e6

#	Time (UTC)	Actor	Event	Details	Hash
12	17:37:27.122	docsentinel	fields_extracted	doc_id=DOC-03, n_fields=9, mean_conf=94.8	fe9abbe777
13	17:37:27.125	docsentinel	fields_extracted	doc_id=DOC-04, n_fields=5, mean_conf=99.0	c2d5c4b3e7
14	17:37:28.859	docsentinel	finding_recorded	finding_id=F-001, status=OBSERVATION, severity=high, confidence=0.9, rule=V1-noise-consistency	d714533da7
15	17:37:28.859	docsentinel	finding_recorded	finding_id=F-002, status=OBSERVATION, severity=high, confidence=0.9, rule=V1-noise-consistency	37852c2eeb
16	17:37:28.859	docsentinel	finding_recorded	finding_id=F-003, status=OBSERVATION, severity=medium, confidence=0.8, rule=V3-glyph-size	25b945926f
17	17:37:28.859	docsentinel	finding_recorded	finding_id=F-004, status=OBSERVATION, severity=high, confidence=0.9, rule=V1-noise-consistency	8c5408df67
18	17:37:28.860	docsentinel	finding_recorded	finding_id=F-005, status=OBSERVATION, severity=medium, confidence=0.8, rule=V3-glyph-size	7f223818d2
19	17:37:28.860	docsentinel	finding_recorded	finding_id=F-006, status=OBSERVATION, severity=medium, confidence=0.8, rule=V4-signature-placement	120b119789
20	17:37:28.860	docsentinel	finding_recorded	finding_id=F-007, status=OBSERVATION, severity=high, confidence=0.85, rule=D1-pdf-font	12f042724d
21	17:37:28.860	docsentinel	finding_recorded	finding_id=F-008, status=OBSERVATION, severity=medium, confidence=0.8, rule=D2-pdf-metadata	f9284cb851
22	17:37:28.860	docsentinel	finding_recorded	finding_id=F-009, status=OBSERVATION, severity=low, confidence=0.5, rule=V4-signature-compare	529169a555
23	17:37:28.860	docsentinel	finding_recorded	finding_id=F-010, status=INCONSISTENCY, severity=high, confidence=0.91, rule=C3-salary-net	0f2789eb6f
24	17:37:28.860	docsentinel	finding_recorded	finding_id=F-011, status=INCONSISTENCY, severity=high, confidence=0.94, rule=C5-bank-closing	9e7e9e96fe
25	17:37:28.860	docsentinel	finding_recorded	finding_id=F-012, status=INCONSISTENCY, severity=high, confidence=0.87, rule=X-dob	0de05672ac
26	17:37:28.860	docsentinel	finding_recorded	finding_id=F-013, status=INCONSISTENCY, severity=high, confidence=0.88, rule=X-income	b1229d0afe
27	17:37:28.861	docsentinel	finding_recorded	finding_id=F-014, status=INCONSISTENCY, severity=high, confidence=0.88, rule=X-income	0d063f8149
28	17:37:28.861	docsentinel	finding_recorded	finding_id=F-015, status=POTENTIAL_MANIPULATION, severity=high, confidence=0.97, rule=R1-convergence	a2669e20ff
29	17:37:28.861	docsentinel	finding_recorded	finding_id=F-016, status=POTENTIAL_MANIPULATION, severity=high, confidence=0.97, rule=R1-convergence	088f946d77
30	17:37:28.861	docsentinel	finding_recorded	finding_id=F-017, status=POTENTIAL_MANIPULATION, severity=high, confidence=0.97, rule=R1-convergence	b9c5560e73